

Computer Systems

COMP1071: Durham University

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Part I

Digital Electronics

1 Circuits

1.1 Transistors

Definition 1.1.1: Transistors are electronically controlled switches that turn on or off when voltage or current is applied to a control terminal.

Definition 1.1.2: A nMOS transistor is a sandwich of layers of conducting and insulating material. The gate is a conducting layer, on top of an insulating layer, on top of the substrate which is a semi-conductor. When its gate g is at a high voltage, current is allowed to flow from source to drain, and this is prevented when the gate is at a low voltage.

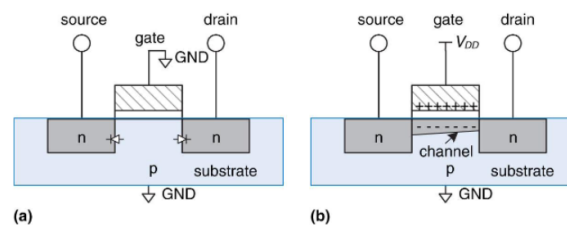


Figure 1: An nMOS transistor

The junction between p-type and n-type silicon is a diode. This means the current can only flow from p-type to n-type as they act as anodes and cathodes respectively.

Definition 1.1.3: A pMOS transistor is the opposite of a nMOS transistor - it allows current to flow when the gate is at low voltages, and prevents current flow at high voltages.

Definition 1.1.4: The static discipline states that if inputs to transistors meet valid thresholds, then the system must guarantee outputs will meet valid output thresholds and will not fall into the forbidden zone (outside of the accepted range for high/low voltages).

Definition 1.1.5: Moore's Law is usually quoted as "transistor density doubles every two years", or "computer processing power doubles every eighteen months".

1.2 Circuit Design

Definition 1.2.3: Abstraction is used when designing circuits to simplify problems by hiding unimportant details, highlighting the actual logic problem to be solved.

Definition 1.2.4: Another design principle used is discipline. Here, design choices are deliberately restricted to make circuits easier to model, design, and combine.

Definition 1.2.5: When designing complex systems, engineers use the three-y's: hierarchy, dividing a system into clear modules and submodules; modularity, using well-defined functions and interfaces for modules; and regularity, encouraging uniformity so modules can be swapped or reused.

1.3 Combinatorial Circuits

Definition 1.3.1: A combinatorial circuit is a circuit where the output depends purely on the current input.

Definition 1.3.2: An adder circuit takes the contents of two registers and adds them together.

- **Definition 1.3.3:** A half-adder circuit takes two one-bit binary inputs and adds them, returning a sum bit, and a carry bit.
- **Definition 1.3.4:** A full-adder circuit takes three one-bit binary inputs (one of which is a carry bit) and adds them, returning a sum bit, and a carry bit.

Definition 1.3.5: A decoder circuit takes an n -bit binary number and decodes it into 2^n data lines.

Definition 1.3.6: A multiplexor (mux) circuit uses a binary number to select one of a set of inputs to funnel to its output.

Definition 1.3.7: A tristate takes in two bits, an input bit and an enable bit. If the enable bit is a one, the input bit is outputted, if not then the output is left floating.

1.4 Sequential Circuits

Definition 1.4.1: A sequential circuit is a circuit which changes its output depending on its input history, alongside the current input.

Definition 1.4.2: A circuit which has two stable output states for a given input is known as a bistable circuit.

Definition 1.4.3: An sr-latch takes in two inputs, set S and reset R . If set has a pulse of one, then the output remains at one, even after the pulse ends. If reset has a pulse of one, then the output is reset back to zero. This is a bistable circuit.

Definition 1.4.4: A d-latch has two inputs, a data input D and a clock input CLK . The latch only changes to match the input at D when the CLK signal is high.

Definition 1.4.5: A d-type flip-flop works similarly to the d-latch, but only allows the stored output to change on the rising edge of the clock signal.

2 Introduction to Logic

2.1 Functionally Complete Sets

Definition 2.0.1: A functionally complete set is a set of boolean operators which can be used to express all possible truth tables by combining members of the set into a boolean expression. For example, $\{AND, OR, NOT\}$, $\{NOR\}$, and $\{NAND\}$ are all functionally complete sets.

Part II

Machine Architecture

3 MIPS

3.1 Introduction to MIPS

Definition 3.1.1: A MIPS (Microprocessor without Interlocked Pipeline Stages) is 32-bit RISC processor used in embedded systems and consumer electronics. It has 32 general purpose registers as seen in

Number	Name	Purpose
0	\$0	Stores the constant value 0.

Table 1: MIPS' 32 general purpose registers

Definition 3.1.2: Amdahl's Law states that the performance improvement to be gained from using some faster mode of execution is limited by a fraction of the time the faster mode can be used.

4 The ATmega328P Microcontroller

5 Programming in AVR Assembly

5.1 Directives

Definition 5.1.1: The assembly code contains statements called directives. These do not correspond to assembly instructions, but the assembler will take them into account when compiling. In AVR assembly, a directive is preceded by a dot. Some examples include:

- `.equ` assigns a value to a label which can be used in later expressions. This is a constant. (`.equ pi, 3`).
- `.globalstart` and `.globalbtnLED` declare the functions `start` and `btnLED` as global.

Part III

Databases

Part IV

Operating Systems

Part V

References

6 Module Professors

The below is accurate as of the 2025/26 academic year and will not be updated in subsequent years.

- *Digital Electronics* and *Machine Architecture* are taught by Ioannis Ivrisimtzis.
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7 Cited Works